SMART INDIA HACKATHON

PROJECT CONCEPT NOTE

DIRECT FARMER-TO-CONSUMER DIGITAL MARKETPLACE

A technology-based approach to reduce unnecessary intermediaries, improve farmer earnings and make agricultural products more affordable for consumers.

Problem **Multiple intermediaries reduce farmers' earnings and increase consumer prices.**
Proposed solution **A digital platform that helps farmers list and sell produce directly to consumers.**
Expected outcome **More transparent pricing, better market access and fewer unnecessary transaction layers.**

Prepared as an initial SIH project concept. The team can customize the title, features and technology according to the selected problem statement and prototype scope.

1. Problem Statement

Farmers often depend on several intermediaries—such as commission agents, wholesalers, distributors and retailers—to move agricultural produce from farms to consumers. Each stage may add handling, transport and profit costs. As a result, the farmer may receive a comparatively low share of the final selling price, while consumers may pay more.

The extent of this issue varies by crop, location, season, transport costs and market conditions. The proposed project aims to improve transparency and provide an additional direct-selling channel; it does not assume that every intermediary is unnecessary.

2. Proposed Solution

Direct Farmer-to-Consumer Digital Marketplace is a mobile-friendly web or mobile application that connects farmers with nearby consumers, shops, restaurants and other buyers. Farmers can list available produce, quantity, quality details and asking price. Buyers can discover products, compare listings, place orders and arrange collection or delivery.

3. Objectives

• Help farmers reach more buyers and explore better selling opportunities.
• Give consumers clearer information about product origin, quantity and price.
• Reduce dependence on unnecessary transaction layers where direct trade is practical.
• Support local markets and improve access to nearby agricultural produce.
• Provide a simple digital interface that can support regional languages.

4. Proposed User Flow

FARMER Registers and verifies profile	MARKETPLACE Lists and displays produce	BUYER Searches and places an order
Adds crop, quantity, location and price	Shows listings, availability and price details	Confirms order and delivery / pickup option

5. Core Features

Farmer registration: Create a farmer profile with contact and farm/location details.
Product listing: Add crop name, photos, available quantity, quality/grade, harvest date and asking price.
Buyer search: Search and filter produce by crop, location, quantity and price.
Price information: Display seller asking prices and, where a reliable data source is available, reference market prices.
Order management: Allow buyers to submit orders and farmers to accept or reject them.
Delivery or pickup: Let both parties select a practical fulfilment option.
Language support: Provide a simple interface in Tamil and other selected languages.

6. System Architecture (Conceptual)

Farmer App / Web Portal **Product upload, stock updates**	**Buyer App / Web Portal** **Search, compare, order**
Application Backend / APIs **Authentication • Listings • Orders • Notifications**	**Database** **Users • Products • Orders • Locations**
Optional Integrations **Maps • Payment gateway • SMS / notifications**	**Admin Dashboard** **Verification • Reports • Support**

Conceptual architecture only. The final architecture should be adjusted to the prototype, security needs and available APIs.

7. Suggested Technology Stack

Layer	Possible technologies
Frontend	HTML, CSS, JavaScript, React or Flutter
Backend	Python (Django / Flask) or Node.js
Database	MySQL, PostgreSQL or Firebase
Maps / location	Google Maps Platform or OpenStreetMap-based services
Notifications	Email, SMS or in-app notifications
Payments	A suitable payment gateway, if included in the prototype

8. Innovation and Differentiation

The project focuses on making local direct trade easier through location-based discovery, transparent seller listings and simple order coordination. Potential enhancements include:

- Tamil-first voice or text assistance for farmers.
- Suggested price ranges using verified market data, with source and date clearly shown.
- Demand insights based on buyer searches and orders.
- Group orders from nearby households or small businesses to make collection more practical.
- Basic quality and seller verification to improve trust.

9. Expected Impact

- **For farmers:** More buyer access, visibility for their produce and an additional selling channel.
- **For consumers:** Better price visibility, local product discovery and clearer seller information.
- **For communities:** Stronger local market connections and improved digital participation in agriculture.

10. Challenges and Mitigation

Challenge	Possible mitigation
Trust and product quality	Seller verification, clear product details, ratings and complaint support.
Delivery and freshness	Offer pickup or local delivery options; show harvest and listing dates.
Digital literacy	Simple screens, Tamil language support and assisted onboarding.
Unreliable price data	Show source and update date; label indicative prices clearly.
Low initial participation	Pilot in one locality with a small group of farmers and buyers.

11. Minimum Viable Prototype (MVP)

For a hackathon demonstration, begin with a small but complete working flow rather than trying to build every advanced feature.

Suggested MVP screens:

1. Welcome / role selection (Farmer or Buyer).
2. Farmer registration and product listing form.
3. Buyer home page with available products.
4. Product details page with seller, quantity and price.
5. Place-order page and farmer order dashboard.
6. Simple admin page to review listings.

Demo scenario: A farmer lists 50 kg of tomatoes at an asking price. A nearby buyer searches for tomatoes, views the listing and places an order. The farmer receives the order and confirms pickup or delivery. The team demonstrates the listing, search and order status changes.

12. How to Measure Success

During a pilot, collect measurable results such as:

• Number of farmers and buyers who successfully register.
• Number of active product listings and completed orders.
• Time taken for a farmer to create a listing.
• Difference between the farmer's received price and the consumer's paid price for comparable transactions, where reliable baseline data exists.
• User feedback on ease of use and trust.

13. Conclusion

The Direct Farmer-to-Consumer Digital Marketplace aims to address the gap between farmers and buyers by providing a simple digital channel for product discovery, transparent listings and order coordination. By piloting the solution in a selected locality and measuring actual outcomes, the project can demonstrate how technology may help farmers access more buyers and help consumers make informed purchasing decisions.

Important note for SIH submission

This document is an initial project concept, not an official SIH problem statement or a claim that the platform has already achieved these outcomes. Before submission, align the idea with the exact SIH problem statement, identify the target users and pilot region, validate the problem through farmer/buyer interviews, and clearly explain what your team will build during the hackathon.

SMART INDIA HACKATHON

Direct Farmer to Consumer Digital Marketplace

Bridging the gap between Farmers and Consumers for a better tomorrow

Fair Prices
Better Income
Healthy Food
Sustainable Future



Farmers
List Products

Farm Fresh Direct
From Our Farmers to Your Home

Consumers
Buy Direct



Problem Statement

Multiple intermediaries reduce farmers' earnings and increase consumer prices.

Farmers often depend on wholesalers, distributors and retailers to sell their agricultural products. Each intermediary adds a profit margin, which reduces farmers' income and increases the price paid by consumers.

Proposed Solution

Smart Farmer-to-Consumer Digital Marketplace

A mobile/web application that connects farmers directly with consumers, removing unnecessary intermediaries and ensuring fair prices for both.

Key Features

- **Farmer Registration** — Farmers can create accounts
- **Product Listing** — Upload crop names, quantities and prices
- **Direct Selling** — Consumers can purchase directly from farmers
- **Live Price Updates** — Display current market prices
- **Location-Based Search** — Find nearby farmers and products
- **Order Management** — Track orders and deliveries
- **Multilingual Support** — Support Tamil and other Indian languages

From Middlemen to Direct Connection

Existing System (with multiple intermediaries)

- Farmer (Sells at ₹10/kg)
- ↓
- Wholesaler (Buys at ₹10 → Sells at ₹15)
- ↓
- Distributor (Buys at ₹15 → Sells at ₹20)
- ↓
- Retailer (Sells at ₹30)
- ↓
- Consumer (Pays ₹30/kg)

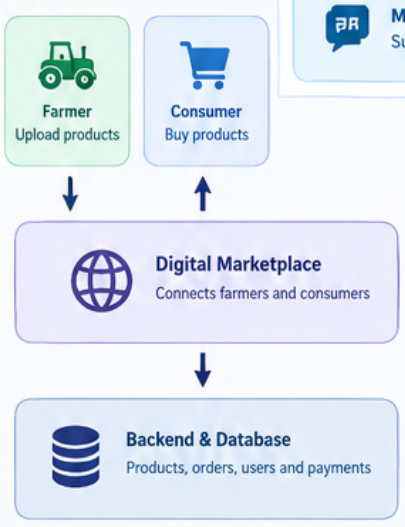
More intermediaries = Higher price

Our Solution (Direct Sale)

- Farmer (List products)
- ↓
- Digital Marketplace (Connects both)
- ↓
- Consumer (Buys at fair price)

Less intermediaries = Better price

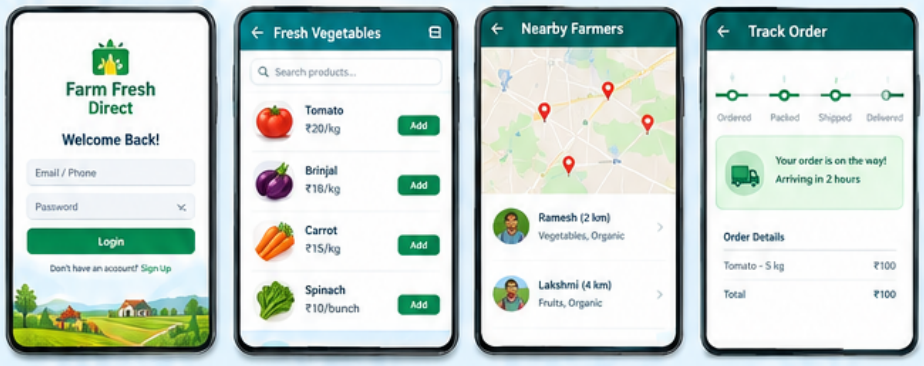
System Architecture



Technology Stack

- **Frontend** — HTML, CSS, JavaScript / React
- **Backend** — Python / Node.js
- **Database** — MySQL / Firebase
- **Maps** — Google Maps API
- **Platform** — Web or Android App

Sample UI



Login / Registration | Product Listing | Location Based Search | Order Tracking

Expected Impact

- Higher earnings for farmers.
- Lower prices for consumers.
- Reduced dependency on middlemen.
- Transparent agricultural pricing.
- Better access to local markets.

Innovation

Use of technology to create a transparent, efficient and farmer-friendly supply chain.

Empowering Farmers | Feeding Communities | Building a Sustainable Future